

Hykabai – Reduced footprint Mobile Robot

A mobile robot capable of transporting goods in reduced spaces, such as a dinner table.

Why mobile robots?

Automation in manufacturing environments has become a necessity to improve both worker security and productivity indicators. Yet, for Small and Medium Companies, the investment required is too big, making it inaccessible for them.

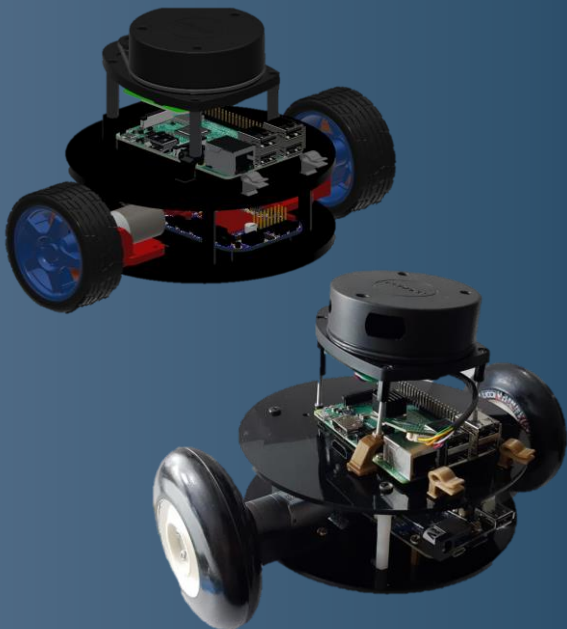
Market

The material transport mobile robot market is expected to reach an \$8.5 billion dollars revenue by 2027, and a quarter of the total mobile robots installed by that date.

As for the entry market, given that this robot's technical characteristics allow the robot to be built, modified and repaired by robotics students, while operating on the standardized Robot Operating System at less than 50% the cost of the competitors.

Financial Overview

Differential robots for Robotics education start around \$500 to \$700 USD. This robot is only \$300.



Characteristics

- It's payload capacity is up to 5kg.
- The robot fits in 0.49 m².
- Uses commercial off-the-shelf (COTS) components, easy to maintain and replace.
- Needs only 10 components to manufacture, and offers a wide range of possible materials.
- Autonomous navigation using SLAM, and remote operation through Manufacturing Execution Software (MES).

Our Achievements

Hykabai is currently implemented in the Microfactory, an automated desktop manufacturing environment for producing prototype Printed Circuit Boards (PCB), which reduces its development time from 7 to 14 to days, down to only 2 days, and it allows other desktop manufacturing devices to be integrated, such as a 3D printer, a laser cutter, and so on. Making it the ideal flexible integrated system for small manufacturing enterprises.

Currently there are two versions already manufactured, and the Mechatronics program at UNAL Sede La Paz is beginning to build its own Hykabai.



Affiliates



Sede Bogotá
Sede la Paz



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Minciencias

